THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 8: **Fractions in Disguise** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two articles are sold at ₹ 1800 each, one at a gain of 10% and the other at a loss of 10%. The overall result is a loss of (in %): 1

(a) 1% **(b)** 10% **(c)** 5% **(d)** 0%

2. The population of a town is 50000 and it grows at 10% per annum. After 2 years it will be: 1

(a) 60500 **(b)** 60000 **(c)** 54450 **(d)** 66550

3. A machine worth ₹ 50000 depreciates at 10% per annum. Its value after 2 years (in ₹) is: 1

(a) 44550 **(b)** 36450 **(c)** 40500 **(d)** 40000

4. A price is first increased by 20% and then decreased by 20%. The net change (as percent, + for increase, − for decrease) is: 1

(a) −4% **(b)** −3% **(c)** 0% **(d)** −5%

5. The CI on a sum for 2 years at 20% p.a. is ₹ 1100. The sum (in ₹) is: 1

(a) 2250 **(b)** 2500 **(c)** 2750 **(d)** 5000

6. A trader marks his goods 40% above cost and allows a discount of 10%. His gain percent is: 1

(a) 26% **(b)** 30% **(c)** 29% **(d)** 23%

7. If the difference between the CI and the SI on a sum for 2 years at 4% is ₹ 12, the sum (in ₹) is: 1

(a) 6750 **(b)** 7500 **(c)** 15000 **(d)** 8250

8. A sum triples in 4 years at simple interest. The rate of interest per annum is: 1

(a) 55% **(b)** 45% **(c)** 100% **(d)** 50%

9. **Assertion (A):** 25% of 360 is 90. 1

Reason (R): $x\%$ of a quantity N equals $\frac{x}{100} \times N$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** A shirt marked at ₹800 is sold at a 10% discount; its selling price is ₹720. 1

Reason (R): Profit or loss percent is calculated on the cost price.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A number is increased by 25% and the result is decreased by 10%. Is the final number more or less than the original? By what percent? 2

12. The cost of an item is ₹ 1500. It is marked 40% above the cost price and sold at 10% discount. Find the selling price. 2

13. Rahul invests ₹ 2500 at 20% p.a. compounded annually. Find the amount after 2 years and write it as a fraction of the principal. 2

14. A number is increased by 40% and the result is decreased by 30%. Is the final number more or less than the original? By what percent? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A merchant sells two items at ₹ 1980 each, one at a gain of 10% and the other at a loss of 10%. Find the overall gain or loss in rupees. 3

16. The value of a car depreciates by 10% every year. If its present value is ₹ 500000, find its value after 3 years. 3

17. A shopkeeper buys an article at ₹ 800, marks it up by 50% and gives a discount of 20%. Find his gain or loss percent. 3

OR

A merchant sells two items at ₹ 2640 each, one at a gain of 20% and the other at a loss of 20%. Find the overall gain or loss in rupees. 3

18. The value of a car depreciates by 20% every year. If its present value is ₹ 400000, find its value after 3 years. 3

OR

A shopkeeper buys an article at ₹ 1200, marks it up by 30% and gives a discount of 20%. Find his gain or loss percent. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Arjun buys a jacket with a marked price of ₹2,500 at a 20% discount. GST of 12% is then added on the discounted price. 5

(i) Find the discount.

(ii) Find the price after the discount.

(iii) Find the GST amount.

(iv) Find the final amount Arjun pays and by how much it is less than the marked price.

20. A sum of ₹ 50000 is invested for 3 years at 10% p.a. compounded annually. Find the interest for each of the three years separately and verify that their total is the compound interest. 5

OR

A shopkeeper marks an article 60% above its cost price. During a sale he gives a discount of 25% and still earns a profit of ₹ 100. Find the cost price. 5

END OF QUESTION PAPER — Total: 40 Marks